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26 February 1974
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MILITARY SPECIFICATION

HITCHES, SNOWPLOW, FRONT-OF-VEHICLE MOUNTING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers snowplow hitches for mounting on military and commercial vehicles.

1.2 Classification. Hitches shall be of the following types and classes, as specified (see 6.2):

Type I - Military, M2 Universal, for 34,000-pound GVW snowplow carrier.
Type II - Military, M2 Universal, for 2-1/2-ton and 5-ton Military trucks.
Class A - Steel.
Class B - Aluminum.
Type III - Commercial.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-C-650	- Crates, Wood, Open and Covered.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-60	- Tape: Packaging, Waterproof.

FSC 3830

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Military

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-I-631	- Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-T-740	- Trucks; 5 Ton, 6x6, Military Design, M39 Series.
MIL-L-2104	- Lubricating Oil, Internal-Combustion Engine, Tactical Service.
MIL-L-10295	- Lubricating Oil, Internal-Combustion Engine, Sub-Zero.
MIL-C-13486	- Cable, Special Purpose, Electrical: Low-Tension, Heavy Duty Single-Conductor and Multiconductor.
MIL-I-15126	- Insulation Tape, Electrical, Pressure Sensitive Adhesive and Pressure Sensitive Thermosetting Adhesive.
MIL-T-21864	- Truck, Lift, Fork: Diesel, Pneumatic-Tire, All Purpose, 10,000-Pound Capacity at 24-Inch Load Center.
MIL-S-52520	- Snowplows, Vehicle Mountings; Straight Blade, Reversible, and V-Blade.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MS27142	- Connector, Plug, Electrical - Pin Contact No. 12, 14 and 16 A.W.G., Waterproof.
MS27143	- Connector, Plug, Electrical: Socket Contact, No. 12 A.W.G., Waterproof.

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MS27144	- Connector, Plug, Electrical: Socket Contact, No. 14 and 16 A.W.G., Waterproof.
MS27145	- Connector, Plug, Electrical: No. 14 A.W.G., Double Contact, Waterproof.
MS27146	- Connector, Receptacle, Electrical: Double Contact, Waterproof.
MS27147	- Adapter, Connector - 3 Way No. 12, 14 and 16 A.W.G., Waterproof.
MS27148	- Contact; Pin, Electrical Connector: No. 12, 14 and 16 A.W.G.
MS33800	- Connector, Receptacle, Electrical - Pin Contact, No. 12, 14 and 16 A.W.G., Waterproof.
MS39020	- Band, Marker, Blank.

DRAWINGS

ME

TA13207E8877	- Hitch, Snowplow, Universal, M2.
TA13207E8888	- Hitch, Snowplow, Universal, M2, Complete W/Side Plates for 2-1/2 Ton, 6x6, M44.
TA13207E8899	- Hitch, Snowplow, Universal, M2, Complete W/Adapter Brackets for 5-Ton, 6x6, M39 Series.
TA13208E9145	- Hitch, Snowplow, Universal, M2, Complete W/Adapter Brackets for 10,000 Lb. Rough Terrain Fork Lift.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, Two Pennsylvania Plaza, New York, NY 10001.)

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NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

B117 - Test for Salt Spray (Fog) Testing.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY

Standard Qualification Procedure AWS B3.0

(Application for copies should be addressed to the American Welding Society, 2501 Northwest Seventh Street, Miami, FL 33125.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., ATTN: Tariff Order Section, 1616 P Street NW, Washington, DC 20036.)

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3. REQUIREMENTS

3.1 Description. Each hitch shall be an assembly consisting of a hitch, vehicle mounting brackets, and a hydraulic system.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The supplier shall furnish one or more hitches for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods will produce hitches that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and unless otherwise specified herein, all examination and tests shall be conducted by the supplier subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified.

3.3 Initial production. When specified (see 6.2), the supplier shall furnish to the Government one or more hitches for inspection as specified in 4.4.

3.4 Materials. Materials shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.4.1 Screw-threads. Threads shall be in accordance with Handbook H28.

3.5 Ground clearance. The minimum ground clearance of the hitches with the vehicle ballasted to rated load shall be not less than 8 inches.

3.6 Instructional material. Two copies of an illustrated list of all parts and two copies of the supplier's maintenance service and operation manual shall be included with each hitch. In addition, instructions shall be furnished for assembling, mounting, and adjusting the hitch and for connecting the hitch headlights and disconnecting vehicle headlights when applicable (see 3.11.5 and 6.8).

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3.7 Identification plates. The hitches shall be identified in accordance with MIL-STD-130. The marking shall be applied to the hitches on plates conforming to MIL-P-514, Type I, Style 1, Composition C of Type I, Grade A, Class 1 material. Vehicle-mounting brackets shall be marked for the vehicle intended.

3.8 Treatment and painting. The portions of the hitches normally painted shall be cleaned, treated and painted in accordance with MIL-T-704, Type A, color as specified (see 6.2).

3.9 Government-loaned property. The following property in the quantity indicated will be loaned by the Government (see 6.2 and 6.4):

Item No.	Description	Identification	Quantity
1.	Truck; any body style, 5-ton, 6x6, M39 Series or M809 Series (dump body style W/O Winch is preferred).	MIL-T-740	1
	or		
2.	Truck; 2-1/2-ton, 6x6, M44 Series.		1
	or		
3.	Truck, Forklift, Rough Terrain, 10,000 Pound.	MIL-T-21864 (FSN 3930-073-8750)	1
	or		
4.	Vehicle, commercial or military (other than as listed) make and model as specified.		1
	and		
5.	Snowplow, vehicle mounting; straight, reversible, or V-blade, as specified.	MIL-S-52520	1

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3.10 Type I hitch. The Type I hitch shall be constructed in accordance with Top Assembly TA13207E8877 and as specified herein.

3.10.1 Hydraulic system. The hydraulic system shall be compatible with and shall connect to the hydraulic system of the vehicle. The components essential for operation of the system shall be furnished by the supplier.

3.11 Type II and Type III hitches.

3.11.1 Type II. The Type II, Class A hitch shall be in accordance with Top Assembly TA13207E8877 and as specified herein. The Type II, Class B hitch shall be in accordance with Top Assembly TA13207E8877, except the material shall be aluminum SAE AA 6061-T6, and as specified herein.

3.11.2 Type III. The Type III commercial hitch shall be a hitch of standard commercial design normally furnished with the size truck and snowplow intended for use. Parts and materials not specified herein shall be those used by the supplier on the standard hitch that he regularly sells on the commercial market. Hitches shall conform to the performance and operational requirements specified herein.

3.11.3 Use. The hitch when used with a compatible snowplow shall withstand without damage, permanent deformation, or malfunction, plowing and travel operations at speeds up to 20 mph on unimproved roads, and up to 35 mph on improved roads and on taxiways and runways of airfields. With the hydraulic system holding the hitch in the lift position, height adjustment shall require not more than one adjustment in each 5 miles of travel.

3.11.4 Hydraulic system. The hydraulic system shall consist of a cylinder or cylinders and all valves, hoses, fittings and piping necessary to connect the cylinders to the hydraulic unit to form a fully operational system. The system shall be capable of operating the snowplow power reversing mechanism as well as the hitch cylinders. A minimum number of fittings, joints, and connections shall be used to minimize back pressure, clogging, and leakage. The hydraulic system shall have a working pressure of 1,500 psi maximum. The hydraulic system shall operate at 2,000 psi for a minimum of 15 seconds without leakage, permanent deformation, or breakage of any component or part. The hydraulic system shall operate with oil conforming to MIL-L-2104, Grade 10 or MIL-L-10295, and shall include a full-flow filter.

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3.11.4.1 Hydraulic piston rods. Sliding surfaces of piston rods shall be plated in accordance with the manufacturer's standard commercial practice with hard chromium and a wall thickness of not less than 0.005 inch. Before being plated, the surface shall be hardened to 50 to 55 Rockwell C. After plating, maximum surface roughness shall be 15 microinches, rms. Special process plating may be proposed and used when approved by the contracting officer, provided the proposed plating furnished corrosion resistance equal to or better than the chromium plating required above when exposed to salt spray in accordance with ASTM B117 for a period of 30 hours.

3.11.4.2 Pumping unit. A pumping unit shall be furnished and shall consist of a 24-volt or 12-volt dc electric motor, as specified (see 6.2), a positive displacement gear pump, a relief valve, mounting pads, and a reservoir, combined into a self-contained unit. Antifriction bearings shall be provided. The relief valve shall be set at 2,000 psi. The pump shall have the capacity to raise the hitch with a compatible snowplow from the lowest position to the highest position in not more than 10 seconds. The pump shall be connected to the motor by a flexible coupling. The reservoir shall have sufficient capacity to operate the hydraulic systems on the hitch and on the reversing plow. The motor shall be connected to the vehicle electrical system with cable conforming to MIL-C-13486, Size 1. The maximum current required to operate the motor under normal operating conditions shall not exceed 130 amperes when the hydraulic system is operating at 1,500 psi and shall not exceed 150 amperes when the system is operating at 2,000 psi. The pumping unit shall be mountable in the vehicle cab within reach of the operator in his normal vehicle driving position and when mounted shall not interfere with any vehicle control. A caution plate conforming to MIL-P-514, Type III, Composition C of Type I, Grade A, Class 1 material shall be attached to the pumping unit in a conspicuous location stating: "Warning - Do Not Operate Continuously More Than 15 Seconds."

3.11.4.2.1 Control valves. A 3-way control valve shall actuate the hitch cylinder(s). A 4-way control valve shall be furnished to operate the snowplow reversing hydraulics. The control valves shall be connected directly to the pump.

3.11.5 Headlight system. When specified (see 6.2), the hitch shall be furnished with two headlight assemblies, and individual right and left 3-wire harnesses to connect the hitch headlight system to the headlight circuits of military and commercial trucks. All hitch headlight components

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and wiring shall be located and guarded to provide protection from damage, breakage, and vibration. The mounting of the lights shall conform to SAE J566 and the lights shall be aimed in accordance with SAE J599. The lights shall be installed above the snowplow head so that the shadow cast by the blade in the raised position shall not decrease the driver's normal visibility of the road. The circuits for high beam, low beam, and ground shall be identified near each terminal by means of marker bands. The wires shall be formed into a harness on each side and shall be secured to the hitch members by support straps. The length of each harness shall be the same. The harness shall not become taut or subject to possible damage or abrasion when the hitch is operated to any position. Routing and components shall be provided to prevent ice and snow spillover from breaking the harnesses. The terminals connecting the hitch shall be compatible with the truck's method of connection. The headlight system shall conform to the requirements and item quantities as specified in Table I for the applicable truck.

Table I. Headlight System Requirements

Item	Description	Requirement	Quantity for* each hitch
1	Headlight housing	Commercial, shockproof	2
2	Wire, low-tension	14 A.W.G., MIL-C-13486, automotive polyethylene low temperature tubing, or SAE J562 oil proof tubing weatherproofed with taped ends.	-
3	Harness insulation	MIL-I-15126 & MIL-I-631	-
4	Cushioned support clamp	Commercial	-
5	Circuit number bands	MS39020	-
	For Commercial-Type Trucks		
6	Sealed-beam light	#6013, 12-v, rough service	2
7	Sealed-beam housing connector	SAE J567	2
8	Truck connector	Mate with socket of the sealed beam connector.	2
	For Military-Type Trucks		
9	Sealed-beam light	PAR 56, #4864, 28-v., Trade No. 4880.	2
10	Headlight housing connector and truck connector.	MS27142 through MS27148, MS33800.	-

*Quantities not indicated shall be as necessary for a complete operating system.

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3.11.6 Vehicle mounting brackets (see 6.2). When specified for use with the M44 series truck with winch, the brackets shall be in accordance with Top Assembly TA13207E8888; with the M39 series truck, Top Assembly TA13207E8899; with the 10,000 pound Rough Terrain Fork Lift, Top Assembly TA13208E9145; with the M809 series truck, commercial mounting brackets shall be designed and furnished for mounting the Type II universal hitch. When the Type III commercial hitch is furnished for use with other military trucks or commercial trucks, commercial mounting brackets shall be furnished. Brackets shall transmit the horizontal plowing forces directly into the truck frame. Brackets shall not be attached to the axle. Unless otherwise specified, (see 6.2), no modifications to a military vehicle chassis shall be permitted, except for removal of the towing shackles, tiedown attachments, and winch. Tire clearance with chains shall be in accordance with SAE J683, with wheels in full steer in both directions.

3.12 Workmanship. The hitches shall be free from defects such as incomplete welds, rust, cracks, and other defects that could impair their operation or serviceability.

3.12.1 Steel fabrication. Steel used in the fabrication of the hitches shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Corners shall be square and true. Flame cutting, using a tip suitable for the thickness of metal, may be employed instead of shearing or sawing. Burned surfaces of flame-cut material shall be ground or machined sufficiently to remove ash and cooling checks. All bends of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly.

3.12.2 Aluminum fabrication. Fabrication of aluminum shall be done by methods that will not cause injury to the metal. Care shall be used in handling to avoid scratching and damage to the edges. Finished edges shall be smooth and free from laminations, kinks, bends, winds, and scratches. Corners shall be square and true. Bends shall be made with metal dies or fixtures to assure uniformity of size and shape. Sharp instruments shall not be used for marking bend lines.

3.12.3 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided, and all bolts, nuts, and screws shall be tight.

3.12.4 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be button-head shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

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3.12.5 Welders and welding.

3.12.5.1 Welder and operator qualifications. All welders and welding operators shall maintain current qualifications appropriate to their job under one of the following qualification standards:

(a) Standard Qualification Procedure of the American Welding Society.
(b) ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

The supplier shall require any welder to retake the tests when, in the opinion of the Government, the work of the welder creates a reasonable doubt as to his proficiency. Tests, when required, shall be conducted at no additional expense to the Government. Recertification of the welder shall be made to the contracting officer only after the welder has taken and passed the required retests.

3.12.5.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter.

3.12.6 Machine work. Tolerances and gages for metal fits shall conform to the limitations specified herein and on the applicable drawings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

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4.1.2 Disassembly inspection, M2 hitch. Failure of any test by the preproduction model or initial production model, M2 hitch shall be cause for disassembly, in the presence of a Government representative, of the hitch to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the hitch. Reassembly with accepted components shall be the responsibility of the supplier.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection of preparation for delivery (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The hitches shall be examined as specified in Column 1 of Table II. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. Performance of the hydraulic system for the Type II hitch without malfunction or failure during the operating tests and performance tests shall constitute an acceptable test of the hydraulic components. In the event of hydraulic malfunction or failure, a mockup of the system shall be bench-tested as specified in 4.6.2.5 to support continued use of the make and model components selected by the supplier.

4.3.2.1 Type I hitch. The Type I hitch shall be tested as specified in 4.6.2. Failure of any test shall be cause for performing the inspection specified in 4.1.2.

4.3.2.2 Type II and Type III hitches. The hitches shall be tested as specified in 4.6.2 using a Government-loaned truck (see 3.9). Failure of any test shall be cause for performing the inspection specified in 4.1.2 for Type II and rejection for Type III.

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4.4 Initial production inspection. When specified (see 3.3), one or more initial production snowplow hitches will be selected at random by the Government from snowplow hitches being produced by production tooling and will be examined as specified in Column 1, Table II and tested as specified in 4.3.2 and 4.3.2.1 or 4.3.2.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production snowplow hitch shall not exclude the remaining snowplow hitches from the quality conformance inspection and acceptance provisions specified in Section 4.

4.4.1 Inspection failure. Failure of an initial production snowplow hitch to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production snowplow hitch(es) and shall be cause for refusal by the Government to continue acceptance of production snowplow hitches until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on snowplow hitches previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all snowplow hitches accepted prior to completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each hitch shall be examined as specified in Column 2 of Table II. Presence of one or more defects shall be cause for rejection.

4.6 Inspection schedule.

4.6.1 Examination. Examination shall be in accordance with Table II.

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Table II. Examination Schedule

Preproduction Initial Prod. and Insp. Comp.	Quality Conformance	Defect	Requirement Paragraph(s)
1	2	3	4
X	-	101. Dimensions not as specified.	3.1, 3.1.1 & 3.11.6
X	X	102. Parts or components missing.	Section 3
X	X	103. Assembly incomplete or incorrect.	Section 3
X	X	104. Connectors not as specified.	3.11.5
X	X	105. Treatment and painting not as specified.	3.8
X	X	106. Identification and other marking not as specified.	3.7
X	-	107. Ground clearance not as specified.	3.5
X	X	108. Hydraulic system not as specified.	3.10.1, 3.11.4 thru 3.11.4.2.1
X	X	109. Light system not as specified.	3.11.5

4.6.2 Tests. Test methods specified herein are those contained in the appendix. Tests shall be conducted in accordance with Table III as follows:

Type I hitch - Items No. 1 and 4.
Types II and III hitches - Items No. 2, 3, 4, and when required 5.

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Table III. Test Schedule

Item	Test	Test Paragraph	Requirement Paragraph
1	Operational (Type I).	4.6.2.1	3.10
2	Mounting (Types II and III).	4.6.2.2	3.11
3	Hydraulic relief pressure (Types II and III).	4.6.2.3	3.11.3, 3.11.4
4	Performance, (all types).	4.6.2.4	3.11.3
5	Hydraulic system (Types II and III) (when required, see 4.3.2).	4.6.2.5	3.11.4

4.6.2.1 Operational. Test the Type I hitch in accordance with Test Method No. 1. Evidence of leakage in the hydraulic system, breakage or permanent deformation of any part or component, or inability to raise and lower the load shall constitute failure of this test.

4.6.2.2 Mounting. Test the Type II and Type III hitches in accordance with Test Method No. 2. Nonconformance to 3.11.3 or 3.11.6, evidence of leakage in the hydraulic system, evidence of breakage or permanent deformation of any part or component, inability to raise and lower the snowplow, or interference between any part of the hitch and a tire shall constitute failure of this test.

4.6.2.3 Hydraulic relief pressure. Test the Type II and III hitches in accordance with Test Method No. 3. Nonconformance to 3.11.3 or 3.11.4 shall constitute failure of this test.

4.6.2.4 Performance. Test the hitch in accordance with Test Method No. 4. Evidence of breakage or permanent deformation of any part or component, or nonconformance of the Type II hitch to 3.11.3 shall constitute failure of this test.

4.6.2.5 Hydraulic system. When required (see 4.3.2), bench-test the hydraulic system components in accordance with Test Method No. 5. Evidence of leakage in the hydraulic system, evidence of internal binding or galling, motor current exceeding the maximum allowable requirements, or inability to complete successfully the specified number of cycles shall constitute failure of this test.

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4.7 Inspection comparison. The Government may select snowplow hitches at any time during the contract production period and subject these snowplow hitches to the examination specified in Column 1, Table II and the tests specified in 4.3.2 and 4.3.2.1 or 4.3.2.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production snowplow hitches.

4.7.1 Inspection failure. Failure of an inspection comparison snowplow hitch to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison snowplow hitch and shall be cause for refusal by the Government to continue acceptance of production snowplow hitches until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on snowplow hitches previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all snowplow hitches accepted prior to the completion of the inspection comparison are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.8 Inspection of preparation for delivery.

4.8.1 Preproduction pack inspection.

4.8.1.1 Examination. Examine the preproduction pack for the defects specified in 4.8.2.3. Presence of one or more defects shall be cause for rejection.

4.8.2 Quality conformance inspection of pack.

4.8.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.2.3 Examination. Samples selected in accordance with 4.8.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

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110. Materials, methods, and containers not as specified for level A. Each incorrect material, method, or container shall be considered one defect.
111. Disassembly not as specified for level A.
112. Unprotected surfaces not coated with preservative as specified for level A.
113. Hydraulic system not preserved and packaged as specified for level A.
114. Electrical wiring not sealed as specified for level A.
115. Maintenance tools not preserved and packaged in accordance with the referenced document for level A.
116. Consolidated packaging not as specified for level A.
117. Packing not as specified for level A.
118. Marking missing, illegible, incorrect, or incomplete for level A or C.

5. PREPARATION FOR DELIVERY

5.1 Preproduction pack. The supplier shall furnish a preproduction pack for examination within the time frame specified (see 6.2) to prove, prior to starting production packaging that the applied preservation, packaging, packing, and marking comply with the preparation for delivery requirements of this specification. Examination shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.6). The preproduction pack may be accomplished utilizing either the preproduction model hitch or a production hitch. If the preproduction model hitch is utilized, any preservation, packaging, and packing shall be removed by the supplier at no expense to the Government, when requested by the Government to facilitate comparison between the preproduction model hitch and production hitches.

5.2 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Disassembly. The lights and brackets shall be removed from the hitch. Further disassembly shall be the minimum necessary to safeguard parts vulnerable to damage and loss. Removed bolts, pins, and washers shall be reinstalled in mating parts to prevent their loss.

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5.2.1.2 Unprotected surfaces. Unprotected exterior metal surfaces of the hitches requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with Type P-1 preservative. The preservative shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.2.1.3 Hydraulic system. The hydraulic fluid reservoir shall be filled with the oil recommended for operation, and the pumping unit, when furnished, shall be actuated to assure coating of all interior parts and surfaces. The pistons shall be fully retracted into the cylinders and secured. The system shall be drained and the openings shall be sealed with caps or plugs as applicable. Any exposed portions of piston rods shall be coated with Type P-11 or P-6 preservative and wrapped or covered with barrier material conforming to MIL-B-121, Type I, Grade A, Class 2. The wrap shall be secured in place with tape conforming to PPP-T-60, Type IV. Openings into the electric motor shall be sealed with tape specified herein. Loose fittings, hydraulic hoses, and the electrical cable shall be packaged together in a close-fitting box conforming to PPP-B-636, W6c, Style optional. The box shall be closed and sealed as specified in the appendix to the box specification for Method V. The pumping unit, when furnished, shall be individually packaged in a close-fitting box conforming to PPP-B-636, W6c, W5c, or V3c, as applicable to the weight. The unit shall be blocked, braced, and cushioned within the box to prevent movement or damage. Box closure and sealing shall be as specified herein.

5.2.1.4 Headlights. The headlights shall be individually packaged as specified in 5.2.1.3.

5.2.1.5 Electrical wiring. Electrical wiring secured to the hitch shall have the quick disconnects sealed with tape conforming to PPP-T-60, Type IV.

5.2.1.6 Instructional material. Instructional material shall be preserved together in accordance with MIL-P-116, Method IC-1 or IC-3.

5.2.1.7 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve the repair parts.

5.2.1.8 Maintenance tools. Maintenance tools shall be preserved and packaged in accordance with the level A requirements of PPP-P-40.

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5.2.1.9 Consolidated packaging. The pumping unit, when furnished, and other components such as the headlights, fittings, loose hydraulic hoses, instructional material, repair parts, maintenance tools, and any other small loose components shall be consolidated packaged together in a close-fitting box conforming to PPP-B-601, overseas type, Style I. The contents shall be blocked and braced within the box in a manner to prevent movement or damage. Box closure shall be in accordance with the appendix to the box specification. Strapping shall not be required.

5.2.2 Level C. Each complete hitch, the repair parts, maintenance tools, and instructional material shall be preserved and packaged to afford protection against deterioration and damage from the supply source to the first receiving activity.

5.3 Packing. Packing shall be level A or C, as specified (see 6.2).

5.3.1 Level A. Each complete hitch, preserved and packaged as specified in 5.2, shall be packed in a crate conforming to PPP-C-650, Style A, Type IV. Blocking, bracing, anchoring, closure and strapping shall be in accordance with the appendix to the crate specification.

5.3.2 Level C. Each complete hitch, preserved and packaged as specified in 5.2, shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in a wooden container complying with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.4 Marking. Packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The Type I hitch is intended for mounting rotary- and military-type displacement plows on 34,000 GVW and larger snowplow carriers. The Type II hitch is intended for mounting commercial displacement plows (MIL-S-52520) on military 5-ton M39 or M809 series trucks; or 2-1/2 ton M44 series, trucks (see items 1 and 2 in 3.9); or on the 10,000-pounds RTFLT. The Type III hitch is intended for mounting commercial displacement plows (MIL-S-52520) on other military trucks and on commercial trucks.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class of hitch required (see 1.2).
(c) Time frame required for submission of preproduction model and number of hitches to be furnished (see 3.2).
(d) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests shall be conducted by the Government and which examination and tests shall be conducted by the supplier (see 3.2).
(e) When initial production inspection is required and the number of snowplow hitches to be furnished, when applicable (see 3.3).
(f) Color of finish coat of paint required (see 3.8).
(g) Make, model, and type of vehicle and type of plow to be loaned by the Government (see 3.9).
(h) Whether the electric motor is to be 24-volt or 12-volt (see 3.11.4.2).
(i) When a headlight system is required for the Type II hitch (see 3.11.5).
(j) Type of hitch and mounting brackets required (see 3.11.6).
(k) When modifications to a military vehicle chassis are permitted (see 3.11.6).
(l) Time frame required for submission of the preproduction pack (see 5.1).
(m) Level of preservation and packaging and level of packing required (see 5.2 and 5.3).

6.3 Preproduction model. Any changes or deviations of production hitches from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish hitches conforming to this specification.

6.4 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.9.

6.5 Test data. The contracting officer should require the supplier or test agency to maintain complete and current test data throughout the testing period and to make such test data available to the Government upon request for review and verification. Any failure occuring during the testing should be reported by the tester immediately upon occurrence to the contracting officer. This should be followed, as expeditiously as possible, by the submittal in writing of complete details of the failure, with pictures as

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appropriate depicting failure, supplier's proposed corrective action, extent of delay anticipated, etc., for resolution and approval. The contracting officer should determine the extent and type of retesting required to assure compliance with specification requirements.

6.6 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the supplier of his obligation to preserve, package, pack, and mark the hitches in accordance with this specification.

6.7 Companion items. Hitches conforming to this specification and snowplows conforming to MIL-S-52520 are companion items. Plows and hitches should be procured together as sets, and mounted on the make and model truck (GFE) intended as the carrier as a part of the contract to assure compatibility.

6.8 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts list, and supplier's maintenance and operation manual to be furnished with each hitch.

6.9 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools) and instructions on shipment of hitches.

Custodians:

Army - ME
Air Force - 84

User interest:

Army - AT

Preparing activity:

Army - ME

Project No. 3830-0148

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APPENDIX

FOR

HITCHES, SNOWPLOW, FRONT-OF-VEHICLE

MOUNTING

Scope: The test methods contained in this appendix are to determine whether hitches procured under this specification conform to the operational requirements set forth.

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TEST METHOD NO. 1

OPERATIONAL

1. TEST FACILITY

The test facility shall consist of an improved or paved surface or shop area of sufficient size to permit placement of equipment and surveillance of test operations.

2. TEST PROCEDURE

(a) Assemble the hitch and mount the hitch on the vehicle or an approved fixture with appropriate hydraulic system.

(1) Attach a compatible snowplow or another approved load to the hitch.
(2) Raise and lower the load not less than 5 times.
(3) Increase the load to not less than 4500 pounds.
(4) Raise and lower the load not less than 25 times.

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TEST METHOD NO. 2

MOUNTING

1. TEST FACILITY

The test facility shall consist of an improved or paved surface or shop area of sufficient size to accomodate operation and surveillance during test operation.

2. TEST APPARATUS

(a) Weights.
(b) Scale.
(c) Compatible snowplow.
(d) Measuring device (tape).
(e) Specified vehicle.
(f) Front chains for vehicle.

3. TEST PROCEDURE

(a) Assemble the hitch and mount on the vehicle.

(1) Install chains on front tires of vehicle.
(2) Ballast the vehicle with rated load.
(3) Attach a compatible snowplow.
(4) Raise and lower the snowplow 25 times.
(5) Turn the vehicle wheels to full right steer. Measure clearance between hitch and tire chains.
(6) Turn the vehicle wheels to full left steer. Measure clearance between hitch and tire chains.
(7) Measure and record the road clearance with the plow raised.

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TEST METHOD NO. 3

HYDRAULIC RELIEF VALVE

1. TEST FACILITY

The test facility shall consist of a secondary gravel road.

2. TEST APPARATUS

(a) Pressure gage.
(b) Timing device.
(c) Vehicle with rated load.
(d) Compatible snowplow.
(e) Ammeter (recording).

3. TEST PROCEDURE

(a) Install pressure gage.
(b) Block pressure relief valve.
(c) Raise the hydraulic pressure to 2000 psig using the pump, hold for not less than 15 seconds and release pressure.
(d) Record pressure and amperes when system is operating at 2000 psig.
(e) Remove block from pressure relief valve.
(f) Raise the hitch with mounted snowplow from the normal operating position to maximum transport position.
(g) After reaching maximum transport position, hold the hydraulic control valve handle in the "raise" position for not less than 15 seconds.
(h) Determine the pressure at which the relief valve operates.
(i) Record ampere reading when the system is operating at 1500 psig.
(j) With the snowplow in the raised position and held in this position by the hydraulic system only, drive the vehicle for a distance of not less than 10 miles over the secondary gravel road at an average speed of not less than 20 mph.
(k) Repeat (f), (g), (h), and (i) above.

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TEST METHOD NO. 4

PERFORMANCE

1. TEST FACILITY

The test facility shall be an area large enough to include the following: (a) improved roads, (b) unimproved or gravel roads, (c) 3 to 18 inches of snow. (Alternate approved course may be used.)

2. TEST APPARATUS

(a) Vehicle with rated load.
(b) Compatible snowplow.
(c) Timing device.
(d) Measuring tape.

3. TEST PROCEDURE

(a) All working lights shall be turned on and left on during all performance tests.
(b) Operate the vehicle with mounted snowplow for not less than 1 hour at an average speed of not less than 30 mph over the improved road plowing snow.
(c) Operate the vehicle with mounted snowplow for not less than 1 hour at maximum safe operating speed over the unimproved or gravel road plowing snow.

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TEST METHOD NO. 5

HYDRAULIC SYSTEM

1. TEST FACILITY

The test facility shall consist of an improved surface or shop area of sufficient size to permit placement and surveillance of required instrumentation.

2. TEST APPARATUS

(a) Test bench with the following:

(1) Relief valve.
(2) Control valve.
(3) Temperature gages and recorders.
(4) Pressure gage and recorder.
(5) Ammeter (recording).
(6) Timer and counter.
(7) Oil cooler.
(8) Additional items to complete the system assembly.

3. TEST PROCEDURE

(a) Install the pumping unit in the hydraulic system that simulates the system on the hitch.
(b) Install pressure gage.
(c) Block pressure relief valve.
(d) Raise the hydraulic pressure to 2000 psig using the pump, hold for not less than 15 seconds and release pressure.
(e) Record pressure and amperes when system is operating at 2000 psig.
(f) Remove block from pressure relief valve.
(g) Cycle the pump 200 times in continuous operation. A cycle shall consist of 15 seconds of operation at a pressure of 1500 psig followed by a 3-minute cooling period.
(h) Record the pressure and amp hours not less than 10 times during test in (g) above.
(i) Repeat (c), (d), and (e) after completion of (g) above.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-H-14435D - HITCHES, SNOWPLOW, FRONT-OF-VEHICLE MOUNTING

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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11 Oct 72 (Rev)

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